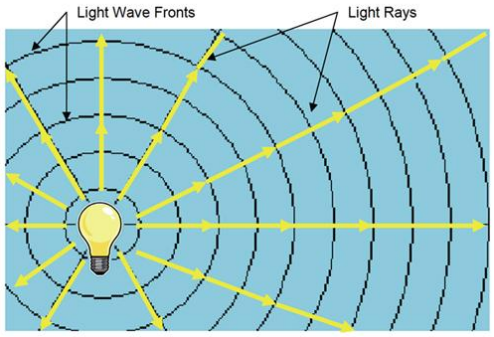
Chapter 57 Introduction to Optics
Calculus Physics II

Optics - is the branch of Physics that studies electromagnetic waves including their generation, behavior, properties, their interaction with materials and other EM waves.

Light as a Wave - When a source emits Electromagnetic waves the waves propagate out as a three-dimensional wave with both $\vec{\mathbf{E}}$ and $\vec{\mathbf{B}}$ oscillating fields. EM waves can be characterized by frequency, wavelength and amplitude of these propagating waves.

Ray Model of Light – Light travels in a straight line from one point in space to another, the lines that the light traces out when traveling between two points can be thought of as rays and can be used to describe the path that the light travels.

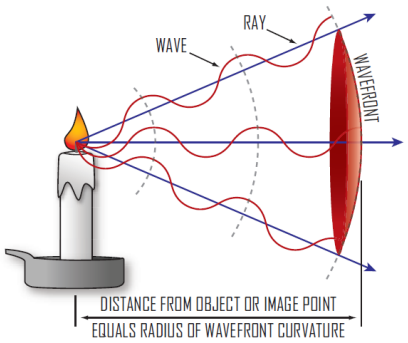
- The ray model of light is useful in describing the path that light travels through optical systems but does not explain EM wave interference and other behaviors of light.
- The human brain assumes light travels in a straight line (a ray) and uses this fact to interpret the world around it, this leads to some misinterpretations of what is actually happening.



Rays – Lines that are perpendicular to the wave front and point in the direction of the velocity of the EM wave and energy flow.

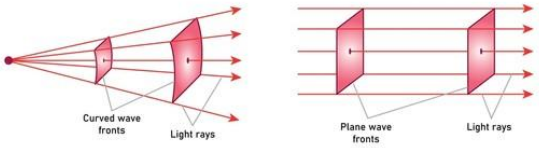
Light travels in a straight-line. Light can be thought of as traveling along the path that is represented by the ray.

- ❖ Rays do not interact with each other.
- ❖ Rays continue on forever unless they interact with matter.



Wave Fronts – Surfaces on which all points of a wave are in the same phase of motion. Example is a point source of light and spherical wave fronts.

Plane wave Fronts (plane waves) – are waves that have flat wave fronts that are not curved. These wave fronts are two dimensional planes that move perpendicularly to the light rays. At large distances from a point source the wave fronts become less and less spherical and approach a flat surface and can be approximated as a plane wave.

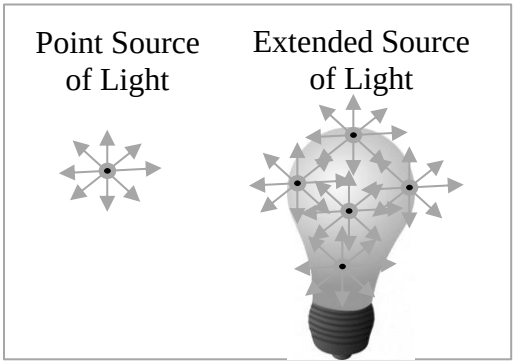


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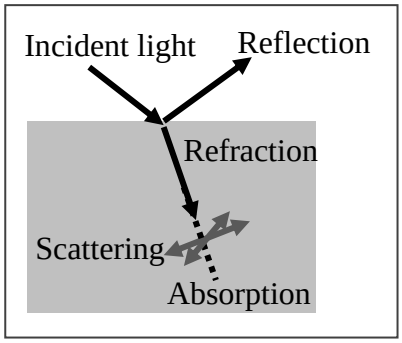
Point sources of light – light appears to be emitted from a single point with no dimensions.

- There are many sources of light that can be concerned points sources when they are a long distance from where they are being observed.

Extended source of light – light that is emitted from a source that is made up of many point sources of light. An extended source is a three-dimensional source of light that has a significant diameter relative to the optical system.



Light interaction with matter – when an electromagnetic wave encounters matter there are a variety of thing that can happen depending on the properties of the matter and what part of the of the EM spectrum the light is composed. When light is incident on matter the following can happen.



1) **Reflection** - the light bounces off the surface.

2) **Refraction** - Light bends as it passes from one material to another. This happens because the speed of light changes when it enters a different medium.

3) **Absorption** - the energy in the light becomes thermal energy in the matter.

4) **Scattering** - the light is emitted in a different direction than the incident light due to the interaction of the light with the particle. This happens when the size of the particle is about the same relative to the wavelength of the light.

5) **Diffraction** - the light bending around corners due to the interference of waves.

What's happening? - When light interacts with matter it is the electric and magnetic fields of the electromagnetic wave that interacts with the material it is having the encounter. The electrons in the material are forced to move (accelerate) due to this interaction, that acceleration of charge causes a EM wave to be generated by the material due to the incident EM wave. This generated wave interacts with the incident wave and depending on how it interacts there can be reflection, transmission, scattering, diffraction or absorption.

- Different materials interact differently with EM waves.
- Different frequencies of EM waves interact with the same material differently.
- Objects of different sizes interact with EM waves differently.
- Different chemical compounds interact with EM waves differently.

Transparent – EM waves moves through the material with minimal absorption, in general you can see through the material.

Translucent – Not all the EM waves are transmitted the material is only partially transparent. Image of the objects looking through the material are blurry or see only as light and dark.

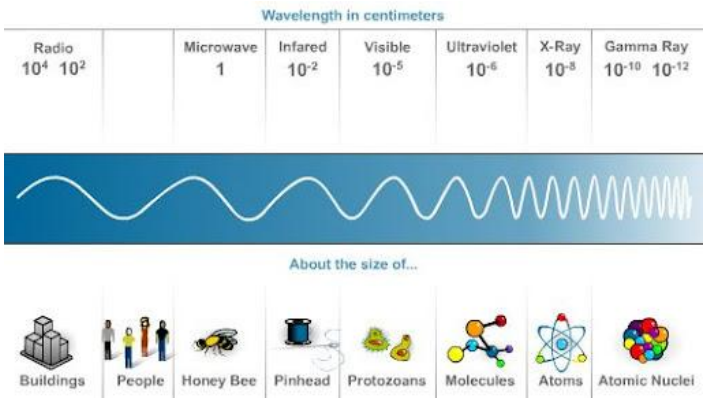
Opaque – The EM wave is absorbed by the material and the energy in the EM wave is converted to thermal energy in the matter. No light is transmitted through the material, all light is absorbed or reflected, cannot see through the material.

Interaction of EM wave with objects - the wavelength (λ) of the EM wave and size of the object (d) For EM waves interacting with matter, what happens to the wave depends on the size of the object relative to the wavelength of the EM wave.

Object much larger than the wavelength of the EM Wave ($\lambda_{EM} >> d_{object}$) – the EM wave reflects, refracts or is absorbed. We see a clear, sharp image of the object from the reflected light off the object. The wave interacts primarily with the surface (reflection, refraction, or casting shadows). This is the basis of ray optics and imaging.

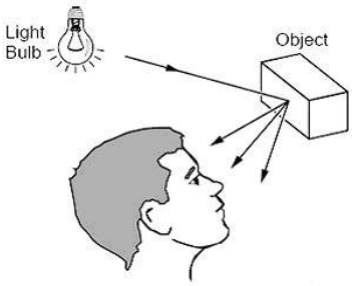
Object has same relative size as the wavelength of the EM Wave ($\lambda_{EM} \approx d_{object}$) - The incoming EM wave is scattered in all directions or diffracted. We see the object as blurry or just a source of light. The wave interacts strongly with the object's structure, this results in significant scattering and diffraction, such as visible light interacting with dust particles or x-rays with crystal lattices.

Object much smaller than the wavelength of the EM Wave ($\lambda_{EM} << d_{object}$) - the incoming EM wave has little to no interaction with the object. The object is "invisible" to the EM wave. low-frequency radio waves pass through buildings, while high-frequency light is blocked.



Reflection of Light – The EM waves "bounces" off the boundary between the two mediums. The light ray changs direction but does not cross the boundary or interface between the two mediums.

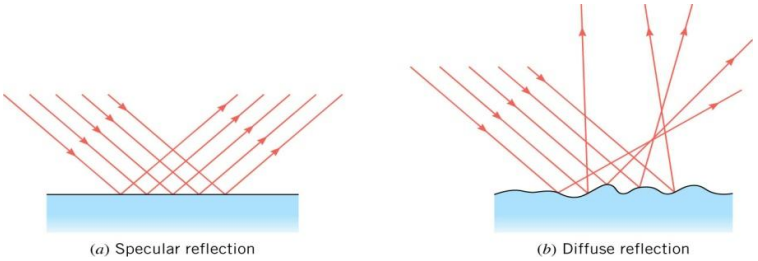
- ✓ In order of an object to be seen, that object must either emit light itself or have light reflected off its surface.



  - Most objects that are visible reflect light.
  - The roughness of the surface will determine the type of reflection and where the light ends up.
  - The "roughness" depends on the wavelength of the EM wave that is being reflected.

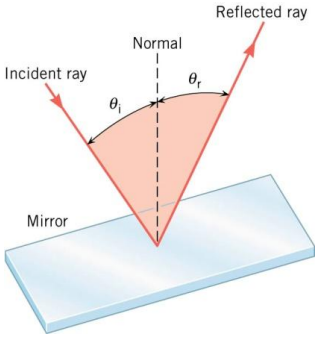
Specular Reflection – When parallel light rays reflect off a surface and remain parallel. Specular reflection creates a visible image from the light of the object.

Diffuse Reflection – Parallel light rays reflect off the surface in many different directions and do not remain parallel. In diffuse reflection, the individual rays still obey the law of reflection. Diffuse reflection does not create an image of the object.

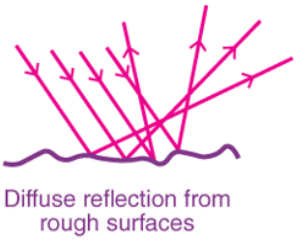


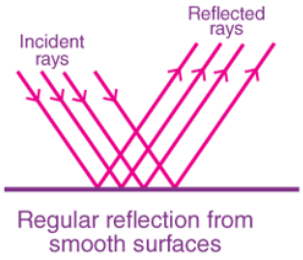
Law of Reflection – For light (EM waves in general) that reflects off a smooth surface, the incident ray, the reflected ray and the normal to the surface all lie in the same plane and the angle of reflection equals to the angle of incidence measured from the line normal to the surface.

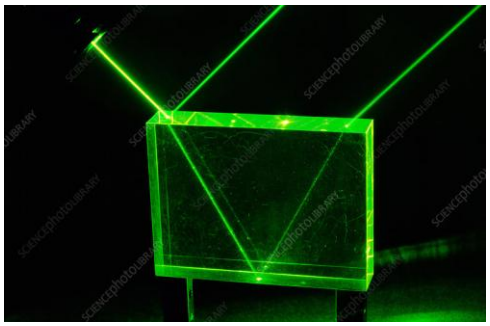
Law of Reflection $\Theta_i = \Theta_r$



- Rough surfaces do obey the Law of Reflection, the angle of the reflected ray is equal to the angle of the reflected ray, but the rays are no longer parallel.
- The light from a green laser is incident on a block of clear glass. Note that at each interface between the air/glass and the glass/air there is reflection. Each reflection obeys the Law of Reflection.

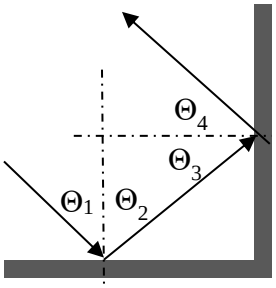






Normal line (the normal) - is a line drawn perpendicular from the surface (boundary between two materials) and is the line that the light rays are measured relative to.

Example: Two flat plane mirrors are placed at right angles, and a ray of light is incident on the bottom mirror at an angle of $\Theta_1 = 40°$, what are Θ_2, Θ_3 and the exit angle Θ_4?



A) Using the law of reflection $\Theta_1 = \Theta_2 = 40°$

B) Using the fact that the angles of a right triangle must sum to 180° and $\Theta_2 + \Theta_3 = 90°$ therefore $\Theta_3 = 90° - \Theta_2 = 50°$

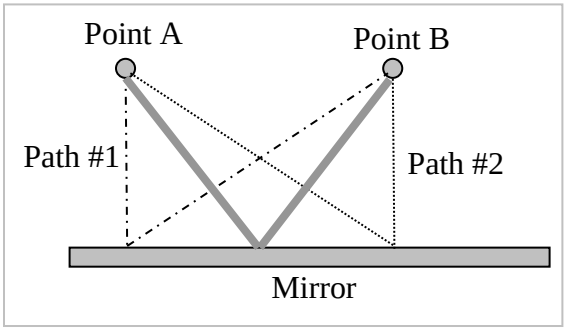
C) Using the law of reflection again $\Theta_3 = \Theta_4 = 50°$

Fermat's Principle for reflection - Light takes the path that requires the least amount of time to travel from point A to point B. Most commonly, the least amount of time is the shortest distance (but not always).

For light going from one point to another the light follows the shortest distance between the two points which is a straight line between them.

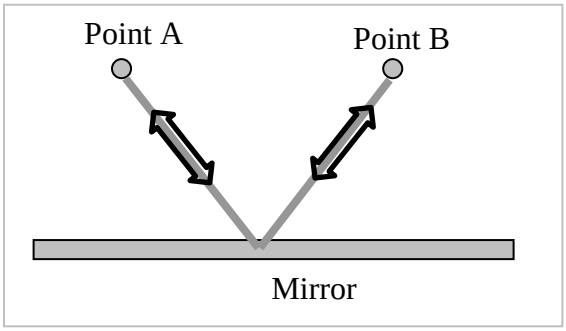


For reflected light the diagram shows two points A and B, when light is emitted at point A and hits the mirror below, for the light to reach point B it will take the path that requires the least amount of time, and it can be shown that the path that is the shortest distance is the bold path and not path # 1 or # 2



<u>Principle of Reversibility of Light</u> – if the direction of the light ray is reversed the light will retrace its original path.

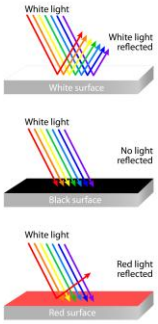
- ❖ The light ray from point A reflects off the mirror and goes through point B, a light ray from point B would reflect off the mirror and goes through point A, the two rays follow the exact path from point A to point B and point B to point A.



<u>Refraction of EM waves</u> – Covered in Chapter 58 – Bending of EM rays due to the change in velocity of the EM radiation.

<u>Absorption of Light</u> – The energy in EM waves is transformed into thermal energy (molecular and atomic vibrations) of the material that has absorbed the waves. The internal energy of the material will increase it will get hotter.

When the natural frequency of vibration of a bound electron in a atom matches the frequency of the light incident on the material that specific frequency is absorbed, the EM wave energy for that frequency is transformed into the motion (vibration) of the electron. For ideal gases the specific frequency's that are absorbed depends on the type of material.





- Materials that are white do not absorb any visible light; all the white light is reflected.
- Black absorbs all visible light.
- Red paper reflects red and absorbs all other wavelengths of visible light.
- Plants absorb EM waves in the blue and red but reflect in the green, that is why leaves look green.
- The absorption of EM waves depends on the material and can be absorbed, transmitted, or reflected.
- We "see" the colors that are reflected NOT absorbed.

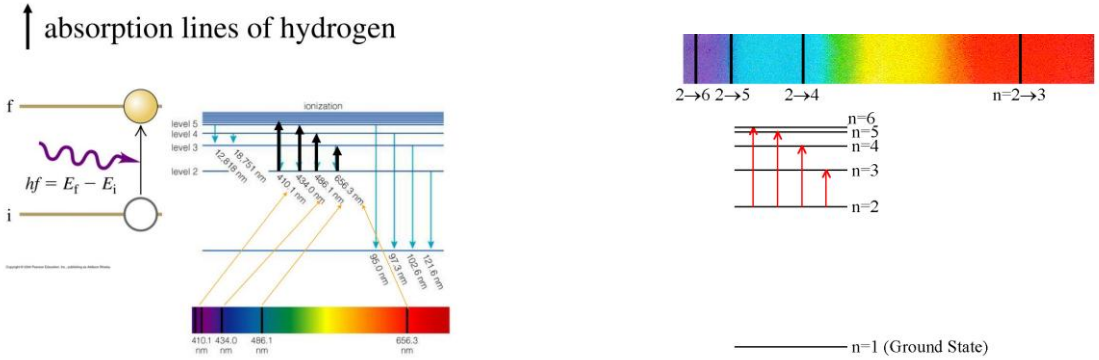
Specific absorption by frequency range of EM waves.

<u>Microwaves</u>: Match the frequency of molecular rotations.

<u>Infrared light</u>: Matches the frequency of molecular vibrations.

<u>Visible and UV light</u>: Can match the natural frequencies of electron clouds in atoms and molecules, causing electrons to jump to higher energy levels.

Absorption by a gas - When white light is passed through hydrogen gas only certain frequencies are absorbed due to the natural frequency of that atom, these appear as black lines in the color spectrum. This is one method of identifying materials by their absorption spectrum.



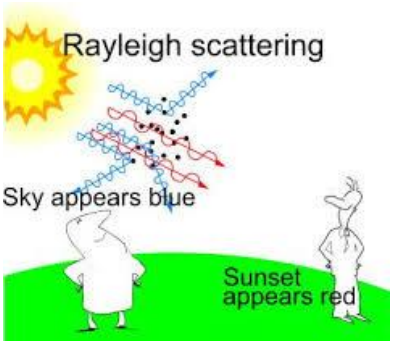
Diffraction of EM Waves – Covered in Chapter 64 – Bending of light around corners or through apertures.

Scattering of light by small particles - Scattering of light is caused by tiny particles in the atmosphere that act like oscillating dipoles or small antenna, when the light is incident on these particles the dipoles start to oscillate at the same frequency as the light due to the oscillating electric field in the light. The oscillating dipole is a charge that is accelerating so it radiates the energy that it has absorbed from the light in different directions.

There are two main types of Scattering Rayleigh and Mie

Rayleigh Scattering – is the scattering of EM waves off particles and molecules that are smaller than the wavelength of the EM wave being scattered. The scattered EM waves are emitted in all directions relative to the incident wave. This type of scattering depends on the frequency of the EM waves that are interacting and is stronger for the higher frequencies. This re-radiated light is radiated along the oscillation direction of the dipole which is perpendicular to the direction of travel of the incident light. This gives the re-radiates light a polarization.

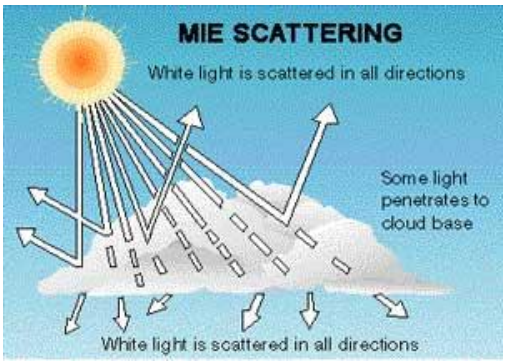
- The scattering of EM waves is dependent on the frequency of the light and is more efficient for the blue light than the yellow or red. So blue light is scattered more, this is why the sky is blue and sunsets are red/orange (no blue/green).



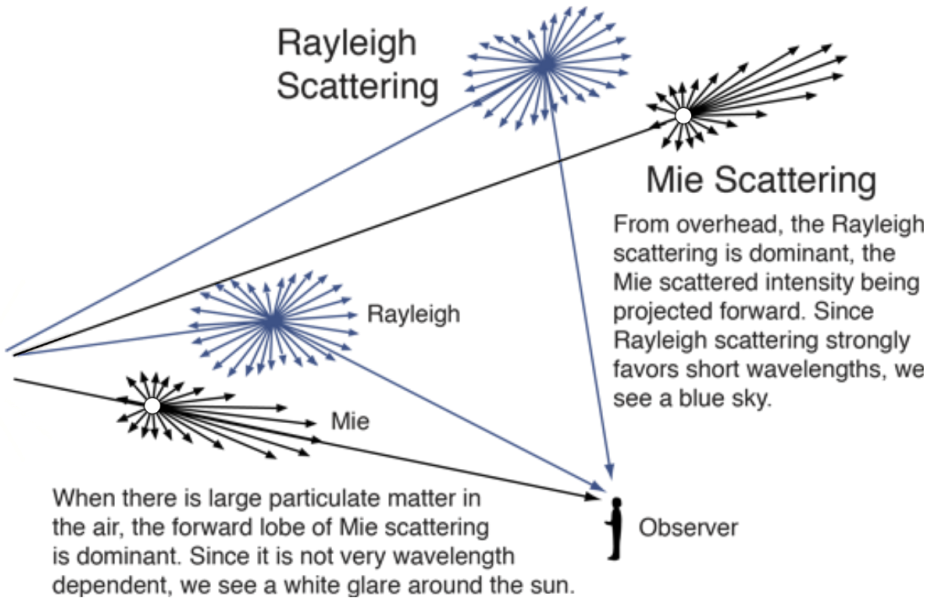


Mie Scattering – scattering of EM waves where the particles are larger than the wavelength of the EM waves. This type of scattering does not depend on frequency.

- For scattering of light from particles larger than the wavelength of the incident light (**Mie scattering**) the scattering does not have the frequency dependence like Rayleigh scattering has but is much more dependent on the shape of the particle. The light scattered from larger particles appear white and only the intensity is diminished. This type of scattering is why clouds appear white because the water droplets in the clouds are much larger than the wavelength of the light.



- Rayleigh Scattering scatters light in more directions and the amount of scattering depends on the frequency of the light relative to the particle sizes.
- Mie Scattering more light is scattered in the direction of the incident light and does not depend on the frequency of light.

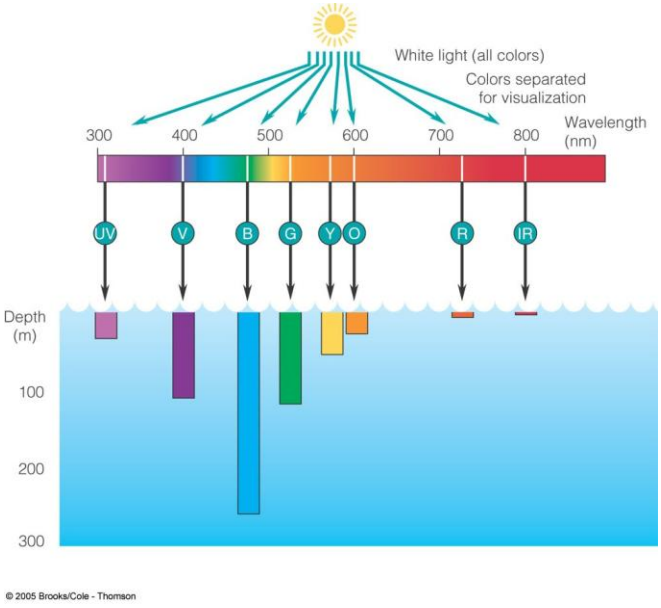


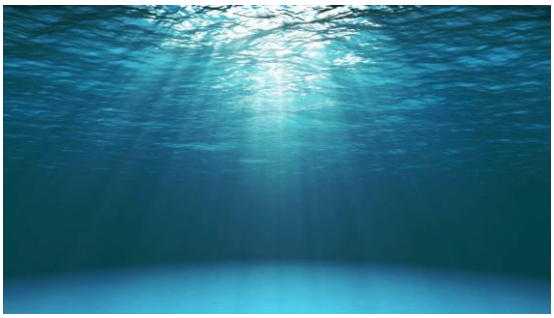
No atmosphere, no scattering or absorption - There must be an atmosphere for scattering of sun light to happen, if there were no atmosphere then the sky would appear black (absence of light) not blue. The moon has no atmosphere therefore the sky appears black except for the earth and stars.



Why is the ocean blue (or blue green)? There are two reasons that water appears blue in large bodies of water.

1) White light from the sun is comprised of all colors but water absorbs the reds, yellow, orange and some green and the water transmits the blue wavelengths, therefore blue light penetrates the water the longest distance (over 100 m deep!) and would reflect off the white sand on the bottom. This is also why deep in the ocean there is no light, almost all light is absorbed after about 100 m in depth (88% of blue light is absorbed).
2) Blue/green light is Rayleigh scattered by the water molecules; the scattering sends the blue light in all directions making the water "look" blue in all directions. The red, orange, and yellow light is absorbed by the water and is not scattered.





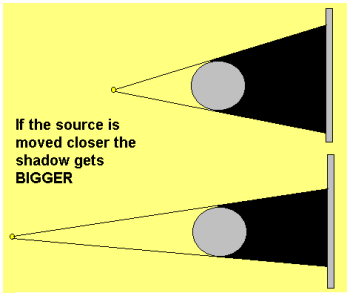
Note: The blue water is NOT caused by the reflection of the blue sky, (although we know that the blue sky is a result of Rayleigh Scattering!)

Shadows – are dark areas due to an object blocking the light ray's incident on that surface.

Dark – is the absents of light.

Shadows from a point source - make well defined shadows.

- The closer the point source of light is to the object that is blocking the light, the larger the shadow.



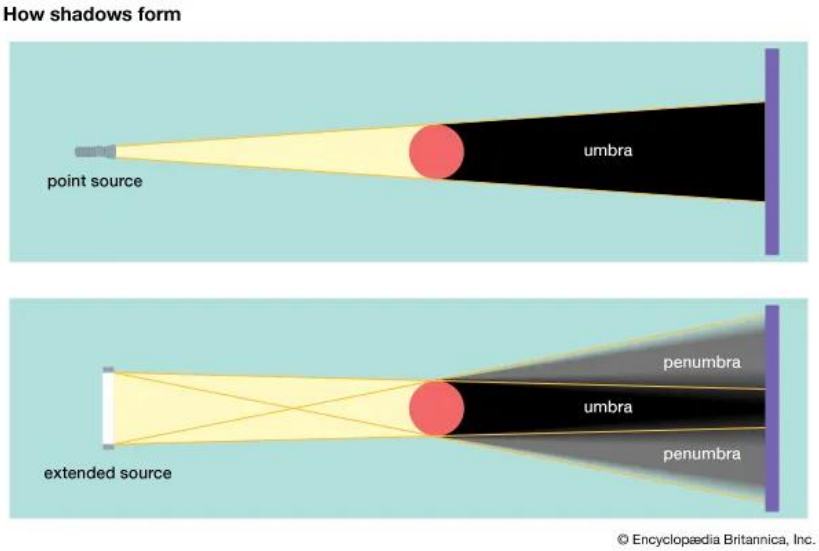


- The farther the light source is from the object creating the shadow the total shadow that is formed is smaller.
- The fist pawn, the light source is far away, and the light is very small compared to the pawn (point source) there is no penumbra, and the shadow is small, the second image of the pawn the light source is much closer, and the light is still a point source compared to the pawn, there is still no penumbra, but the shadow is larger.

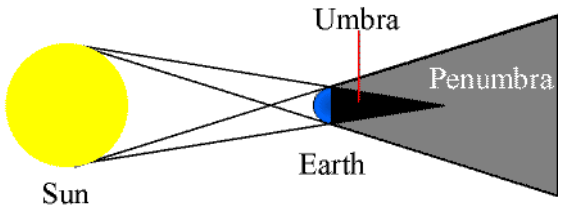
Shadows formed by an extended source – the object only partially blocks the light from a portion of the source therefore there is not a well-defined shadow but a burring of the edges.

Umbra – is the full or perfect shadow (no light)

Penumbra – is the area of the soft shadow not completely dark (gray area)



The Earth's shadow caused by the sun creates an Umbra and Penumbra



- Parallel light (light source far away) will give a perfect shadow with little to no Penumbra, as the source gets larger compared to the object, this causes the Penumbra to form and will get larger as the light source gets closer to the object.

